

Fundamentals of Programming Languages

Exercises 3

SASyLF

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Exercises in SASyLF The following exercises are based on file `tapl-nats.slf`, available at the course's web site, and on the proofs in TAPL.

1. Replace all occurrences of **proof** by **unproved** by appropriate proofs.
2. Exercises 2 introduced the term $t_1 \max t_2$.
 - (a) Add the new term to the language of terms (remember to declare `max` as a terminal).
 - (b) Add the evaluation rules and the typing rule from Exercises 2 (questions 8 and 9). Remember that we want a left-to-right semantics: in the term $t_1 \max t_2$, we first evaluate t_1 and, only once it is a numerical value nv_1 , do we evaluate t_2 .
 - (c) Extend the proofs of theorems preservation and progress, and of lemma uniqueness-**of**-types, to account for `max`.
3. Now consider two new terms, `even t` and `odd t`, whose one step evaluation rules are, in English, as follows:
 - `0` is even,
 - `succ t` is even if `t` is odd, and
 - `succ t` is odd if `t` is even.

State and prove the following two results in SASyLF.

- (a) The successor of the successor of an odd term is odd.
- (b) The successor of the successor of the successor of an even term is odd.

Implementation The following exercises can be based on the Haskell source code adapted from TAPL and provided at the course's web site.

4. Add terms `even` and `odd` to the datatype of terms.
5. Add the corresponding one step evaluation rules to function step.
6. Now add the typing rules to function typeof.